

AN2DL - Second Challenge Report

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1 Introduction

We address a 4-class histopathology classification task (Luminal A, Luminal B, HER2(+), Triple negative) from low-magnification WSI patches, each associated with a binary mask. Our methodology followed a three-step pipeline:

1. Dataset cleaning, preprocessing, and curation;
2. Training of a custom CNN from scratch;
3. Adoption of advanced architectures via **transfer learning** and **fine-tuning**.

2 Problem Analysis

The dataset includes **1,272 image-mask pairs** with heterogeneous resolutions: **691 labeled** for training and **477 unlabeled** for testing. Global re-sizing can remove localized multi-scale cues typical of histopathology images [1].

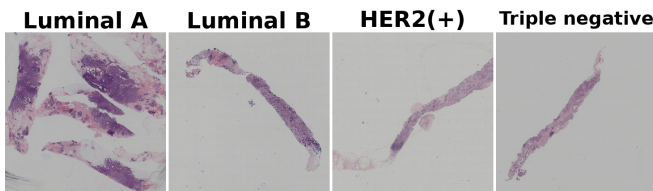


Figure 1: Sample images from each molecular subtype class in the training set.

Quality control. We automatically ranked samples using simple color/texture statistics and manually inspected outliers, a common practice to reduce the impact of dataset noise [5]. This revealed **110 contaminated images**, which were removed. Keeping them caused unstable optimization and worse validation behavior.

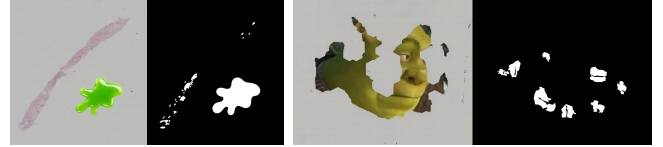


Figure 2: Examples of anomalous samples removed from the training set during quality control.

Spatial sparsity. Relevant tissue occupies a limited image area, motivating mask-guided preprocessing and patch-based modeling.

Class imbalance. Dataset imbalance was handled using class weights to penalize underrepresented classes.

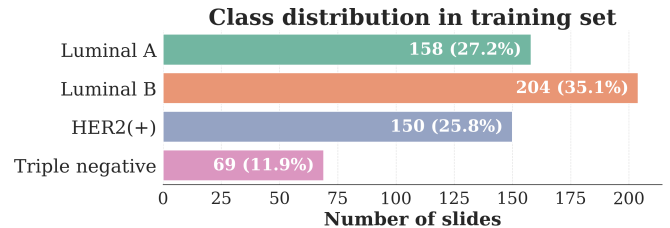


Figure 3: Class distribution in the cleaned training set after removing contaminated samples.

3 Method

3.1 Baseline: CNN from scratch

We started with a small CNN trained from scratch to establish a baseline. Training from scratch on limited medical data is often prone to overfitting [6]. As expected, the model tended to overfit and was sensitive to augmentation: when augmentations were too strong, training often failed to converge or the model collapsed to predict only a few classes; when augmentations were too weak, the model rapidly overfit. Typical scores were ≈ 0.35 **weighted F1 on validation** and ≈ 0.28 **on test**.

3.2 Transfer learning: backbone search

Given the limited size of the labeled set, we shifted from training from scratch to **transfer learning**, i.e., initializing the feature extractor with **ImageNet-pretrained** weights and fine-tuning on the target task. This choice is well-motivated in medical imaging, where low-level and mid-level visual primitives (edges, textures, color patterns) can transfer, while higher-level layers adapt to domain-specific morphology [6].

- We first explored **EfficientNet-B0/B1** due to their strong accuracy/efficiency trade-off and rapid iteration speed [8]. In our setting, they trained reliably but tended to plateau early, suggesting a capacity limit when the model must discriminate subtle histological patterns.
- The `tf_efficientnet_b1_ns` variant was generally more stable (smoother validation curves), likely due to improved pretraining and normalization choices in the *Noisy Student* pipeline.
- We also tested `tf_efficientnetv2_s.in21k`, where larger-scale pretraining improved convergence and slightly increased validation performance [9].

Nevertheless, EfficientNet-based runs remained sensitive to the foreground/background ratio and tended to underutilize fine-grained tissue details when trained with global resizing. These limitations motivated the transition to architectures explicitly designed to better capture localized patterns under patch-based learning.

3.3 Final pipeline: ConvNeXt + mask-guided patching

Our best-performing setup combined three components (implemented in the final notebook) :

1. **Mask-guided crop + background suppression.** We crop each image around the mask bounding box with padding, optionally applying the mask to reduce background influence.
2. **Patch-based training/inference.** To preserve resolution while controlling memory, we avoid global resizing: we sample **one random patch** per image in the training phase, while during inference we extract a **dense grid of overlapping patches** and aggregate predictions [3].
3. **Backbone: ConvNeXt-Base pretrained on ImageNet.** ConvNeXt modernizes CNN design while retaining convolutional inductive bias; in our case it paired well with patching and delivered the most consistent gains [4].

3.4 Training and model selection

We trained with **5-fold stratified cross-validation** to obtain robust estimates and reduce variance. Key choices (final configuration):

- Optimizer: **AdamW**, with **two learning rates** (smaller for backbone, larger for classifier head).
- Loss: **weighted cross-entropy** (fold-wise class weights) + **label smoothing** (0.05). [7]
- Regularization: early stopping on validation F1; mixed precision on GPU.

Table 1 summarizes the estimated validation performance of the evaluated backbones.

3.5 Ensemble inference (submission)

We trained one model per fold and ensembled them at test time. For each test image we aggregate: (i) patch predictions (dense grid), (ii) simple TTA flips, (iii) fold ensemble averaging. To reduce noise, we keep only the **Top-K most confident patches** for each view before combining probabilities.

Table 1: Estimated validation performance of the evaluated backbones. Metrics are reported as averages over cross-validation folds. All results were obtained through fine-tuning. Best results are highlighted in **bold**.

Model	Accuracy	Precision	Recall	ROC AUC
EfficientNet-B0 (Google)	0.33 $\pm$ 0.04	0.34 $\pm$ 0.03	0.33 $\pm$ 0.04	0.62 $\pm$ 0.05
EfficientNet-B1 (Google)	0.35 $\pm$ 0.03	0.36 $\pm$ 0.03	0.35 $\pm$ 0.03	0.65 $\pm$ 0.04
tf_efficientnet_b1_ns	0.38 $\pm$ 0.02	0.39 $\pm$ 0.02	0.38 $\pm$ 0.02	0.69 $\pm$ 0.03
tf_efficientnetv2_s.in21k	0.40 $\pm$ 0.02	0.41 $\pm$ 0.04	0.40 $\pm$ 0.03	0.72 $\pm$ 0.03
ConvNeXt-Base (patch-based)	0.43 $\pm$ 0.02	0.43 $\pm$ 0.02	0.43 $\pm$ 0.02	0.76 $\pm$ 0.03

4 Failed Experiments

Some ideas were promising but did not translate into better validation stability or final scores:

- **Mask as explicit input (dual-branch networks).** We explored dual-branch architectures processing RGB images and masks separately with late feature fusion. This substantially increased model complexity and overfitting, while providing no consistent gains. Masks proved more effective when used as a *preprocessing prior* rather than as a learnable input.
- **Multiple Instance Learning (MIL) with attention.** Images were modeled as bags of patches aggregated via attention mechanisms. Although theoretically appealing, training was computationally expensive and highly sensitive to patch sampling and hyperparameters, resulting in unstable validation behavior and no consistent improvement over simpler aggregation strategies [2].
- **Over-aggressive augmentation.** Strong geometric and color perturbations often disrupted optimization, leading to flat training curves or class collapse. A moderate augmentation regime combined with patch-based sampling yielded significantly more stable and robust learning.

5 Results and Discussion

A major practical challenge was the **high variance of validation F1** across epochs and data splits. This behavior was expected given the limited dataset size, class imbalance, and intrinsic stochasticity of the pipeline (random patch sampling, data

augmentation, and GPU non-determinism). In several runs, transient overfitting was observed, with steadily improving training metrics while validation F1 exhibited noticeable oscillations.

To mitigate these effects, we adopted **5-fold stratified cross-validation** for model selection, combined with early stopping on validation F1 and **fold ensembling** at inference time. Although validation performance was not perfectly stable, this strategy substantially reduced extreme fluctuations and yielded more reliable predictions compared to single-split training.

Overall, the transition from global image processing to **mask-guided cropping and patch-based inference** produced the most consistent performance gains, particularly by reducing background-induced errors and limiting overfitting. Residual errors were primarily associated with ambiguous morphology or extremely small informative tissue regions.

6 Conclusion

This project highlights the central role of data quality, spatial focus, and resolution preservation in histopathology image classification. Initial experiments exposed the limitations of both simple CNNs trained from scratch and overly complex modeling choices under limited supervision. By removing contaminated samples, exploiting masks as preprocessing priors, adopting patch-based full-resolution learning, and combining cross-validation with ensembling, we derived a robust and stable classification pipeline. These results emphasize that **carefully aligned architectural and data-centric choices are often more effective than increasing model complexity alone**.

References

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